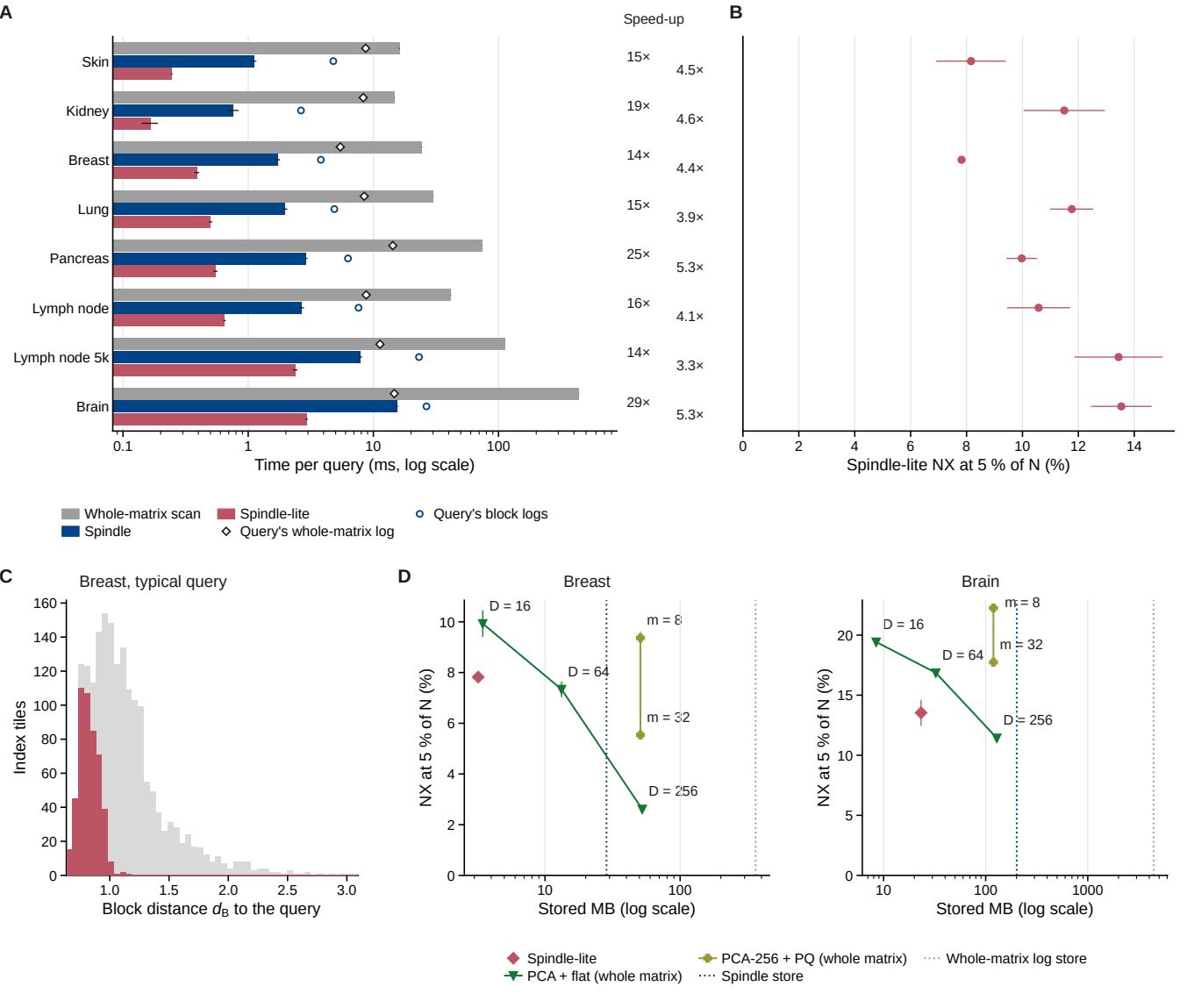

A
Speed-up
Skin
Kidney
Breast
Lung
Pancreas
Lymph node
Lymph node 5k
Brain
15×
19×
14×
15×
25×
16×
14×
29×
4.5×
4.6×
4.4×
3.9×
5.3×
4.1×
3.3×
5.3×
0.1
1
10
100
Time per query (ms, log scale)
B
0
2
4
6
8
10
12
14
Spindle-lite NX at 5 % of N (%)
Whole-matrix scan
Spindle
Spindle-lite
Query's whole-matrix log
Query's block logs
C
Breast, typical query
Index tiles
Block distance d_B to the query
D
Breast
Brain
NX at 5 % of N (%)
Stored MB (log scale)
D = 16
D = 64
D = 256
m = 8
m = 32
Spindle-lite
PCA + flat (whole matrix)
PCA-256 + PQ (whole matrix)
Spindle store
Whole-matrix log store